

Arduino Servo Motor Control

Building Smart Gates





How Does a Toll Gate Lift Up?

90° Exact angle the barrier lifts —
every single time!

How does the barrier lift exactly to
90° and stop there? 🤔

Let's find out with Arduino and
servo motors!

What is a Servo Motor?



Precise Angle Control

A servo motor moves to a precise angle (0° to 180°) and holds its position firmly. You can command it to stop at exactly 45° , 90° , or any angle in between!

Unlike a DC Motor

A DC motor spins continuously and cannot hold a specific angle. A servo motor is smarter — it knows exactly where to stop and stays there!



Meet the SG90 Servo Motor

01

Power & Ground Wires

Red wire = VCC (Power) — connects to 5V on Arduino. Brown or Black wire = GND (Earth) — connects to Ground. Think: Red is power, brown/black is earth!

02

Signal Wire

Orange or Yellow wire = Signal (Control) — connects to PWM pin 9. Mnemonic: Red = power, brown/black = earth, yellow/orange = signal — just like a traffic light!



Wiring Guide:
Red = 5V Power
Brown = GND
Yellow = Pin 9 Signal

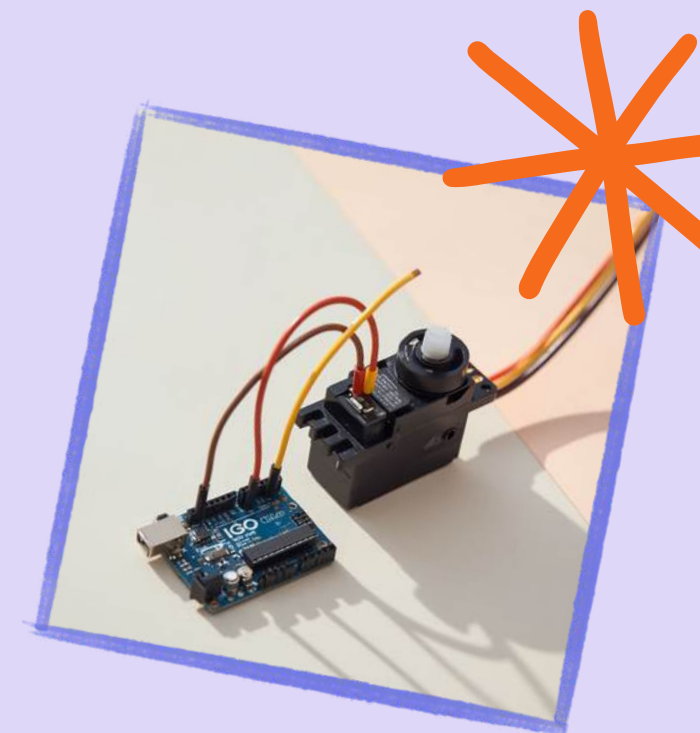


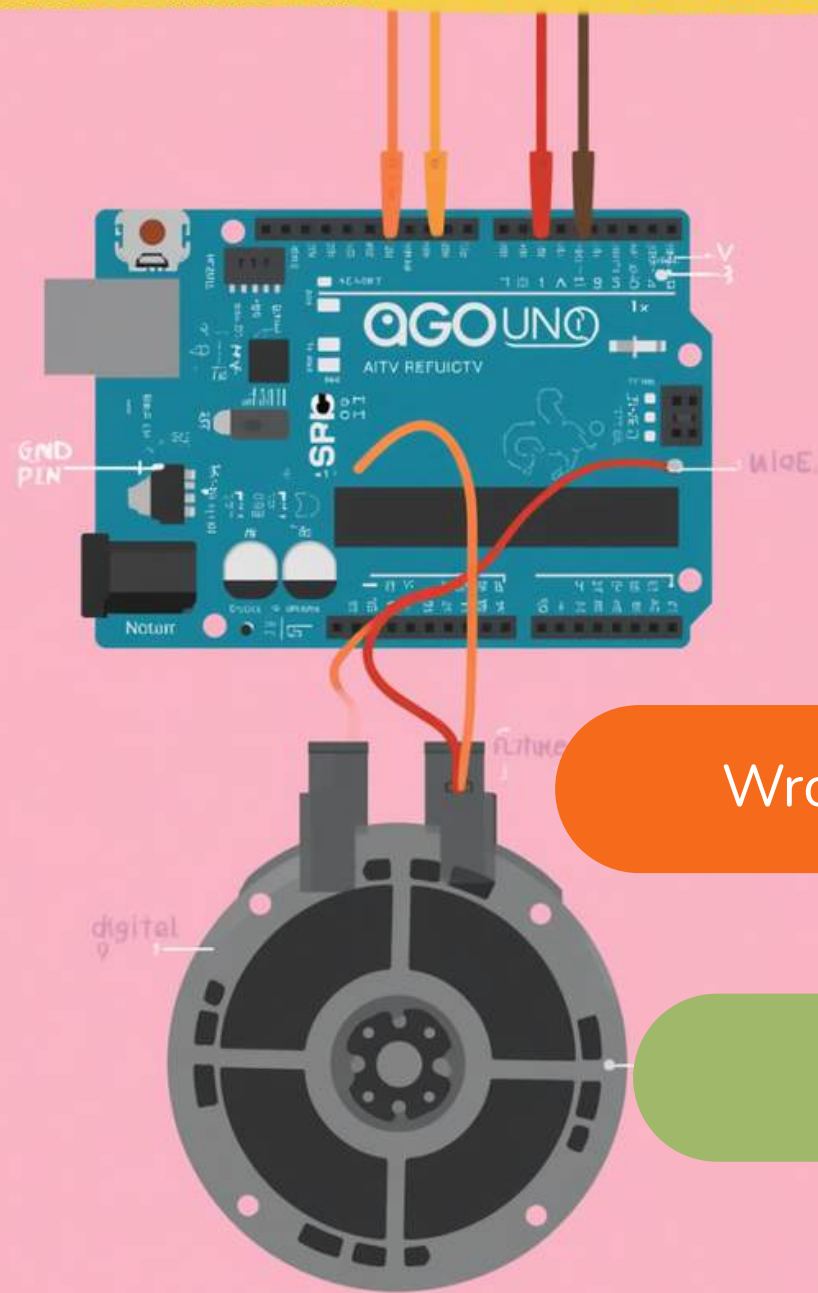
Connecting the Servo!

Connect the SG90 servo to your Arduino UNO using 3 jumper wires:

- Red wire → 5V (Power)
- Brown/Black → GND (Ground)
- Yellow/Orange → Pin 9 (Signal)

Make sure all connections are secure before uploading your code!





Wrong wiring!

Spot & fix it!

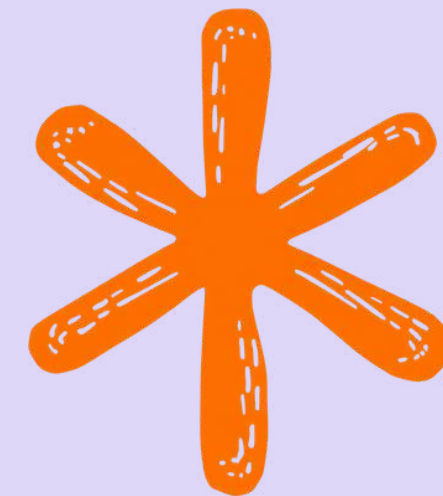
Spot the Wiring Mistake!



Can you find the errors?

Look carefully at the wiring diagram. Two mistakes are hidden! Hint 1: Check which wire goes to pin 9. Hint 2: Is the red wire in the right place? After you find them — reveal the correct wiring below!

Writing Code to Control the Servo



Include the Servo Library

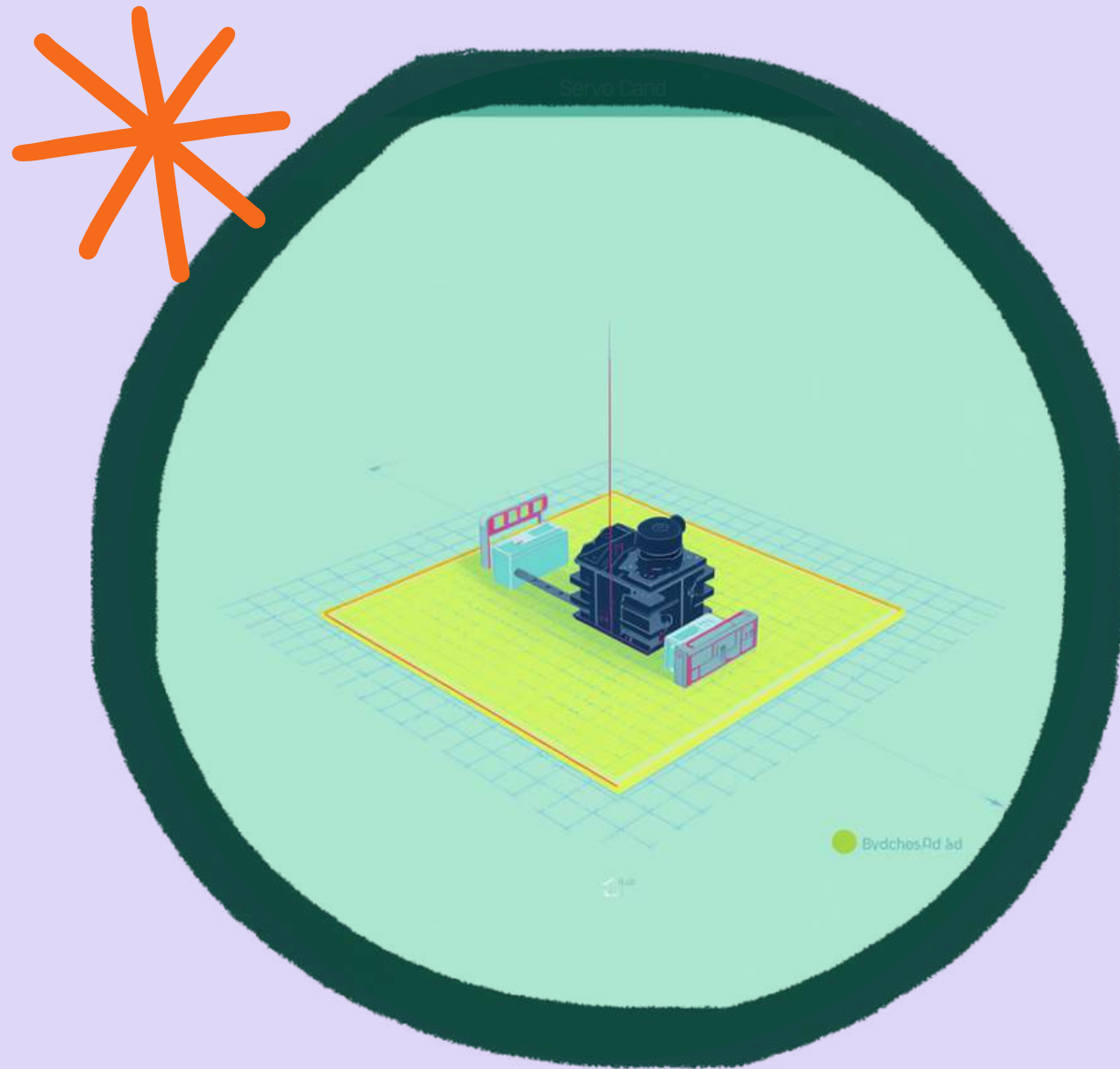
```
#include <Servo.h>
Servo myServo;
void setup() {
  myServo.attach(9);
}
```

Attach & Write Angle

```
void loop() {
  myServo.write(90);
  delay(1000);
}
```

Pin 9 controls the servo signal angle!





Simulating Servo

Movement!

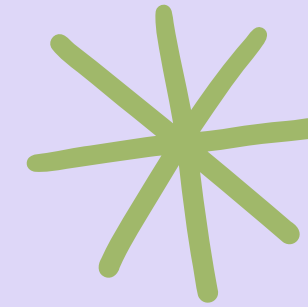
0°→180° Servo sweeps smoothly and returns to start.

The servo sweep code moves the motor from 0° to 180° and back to 0° in a loop. We use `delay()` to pause the servo at each angle — giving it time to reach its position before moving again. Try changing the delay value to make it faster or slower!

Real-World

Examples in India!

Servo motors are used all around us! From toll gates to space — they control precise movements every day.



FASTag Toll Barriers

Lifts exactly 90° to let cars pass!



Society Boom Barriers

Opens & closes at apartment gates

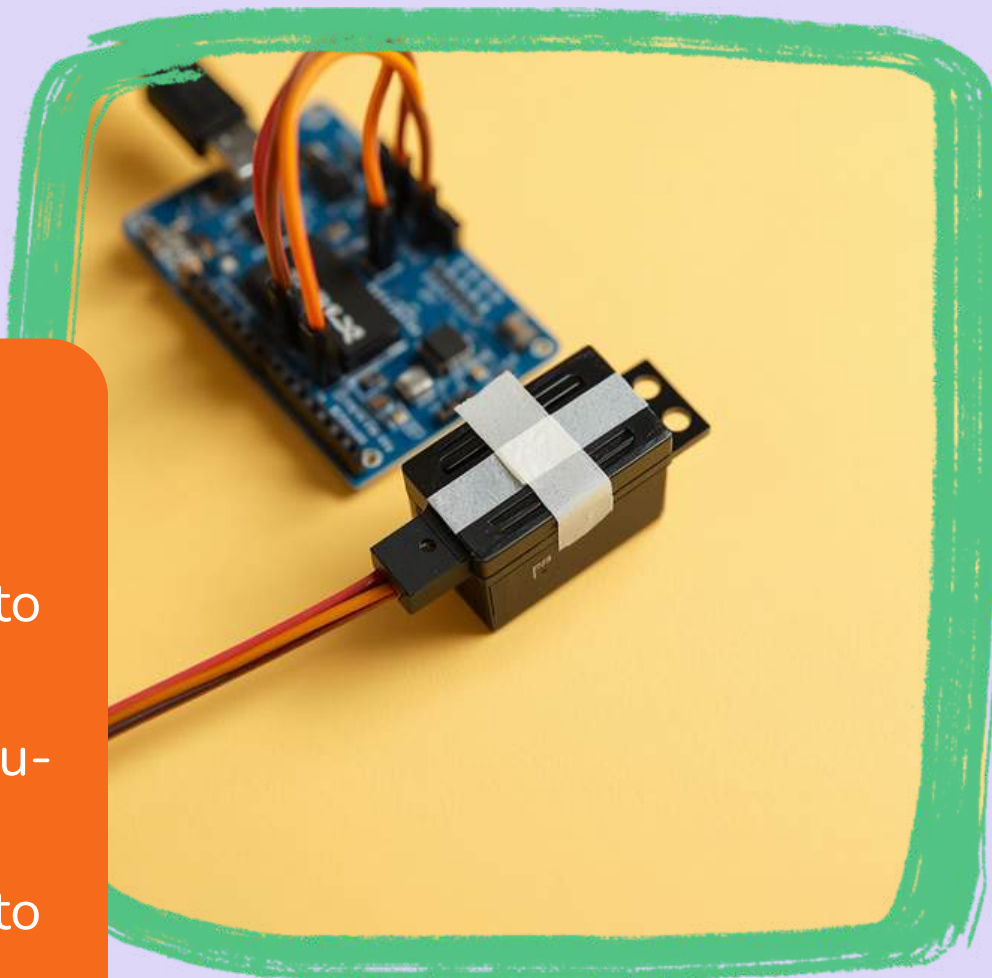


ISRO Satellite Panels

Solar panels deploy using servos!



Attach a popsicle stick to your servo arm using double-sided tape or Blu-tack. Connect your Arduino and get ready to build!



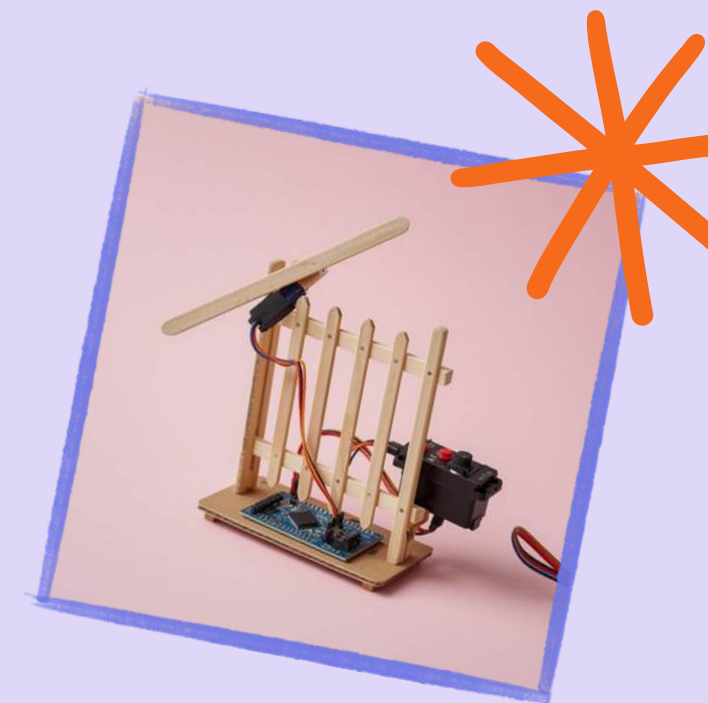
DIY Challenge:

Build Your Smart Gate Arm!

🔧 Use double-sided tape or Blu-tack to fix the popsicle stick to the servo arm.

📐 Code the servo to open the gate to 90° and close back to 0°.

🏠 Try it at home or in class — show us your smart gate!



Let's Build & Code! 🚀

Quiz: Test Knowledge!

Q1

How many wires does an SG90 servo have?

Name all three wire colours and what each one does. Hint: think about power, ground, and signal!

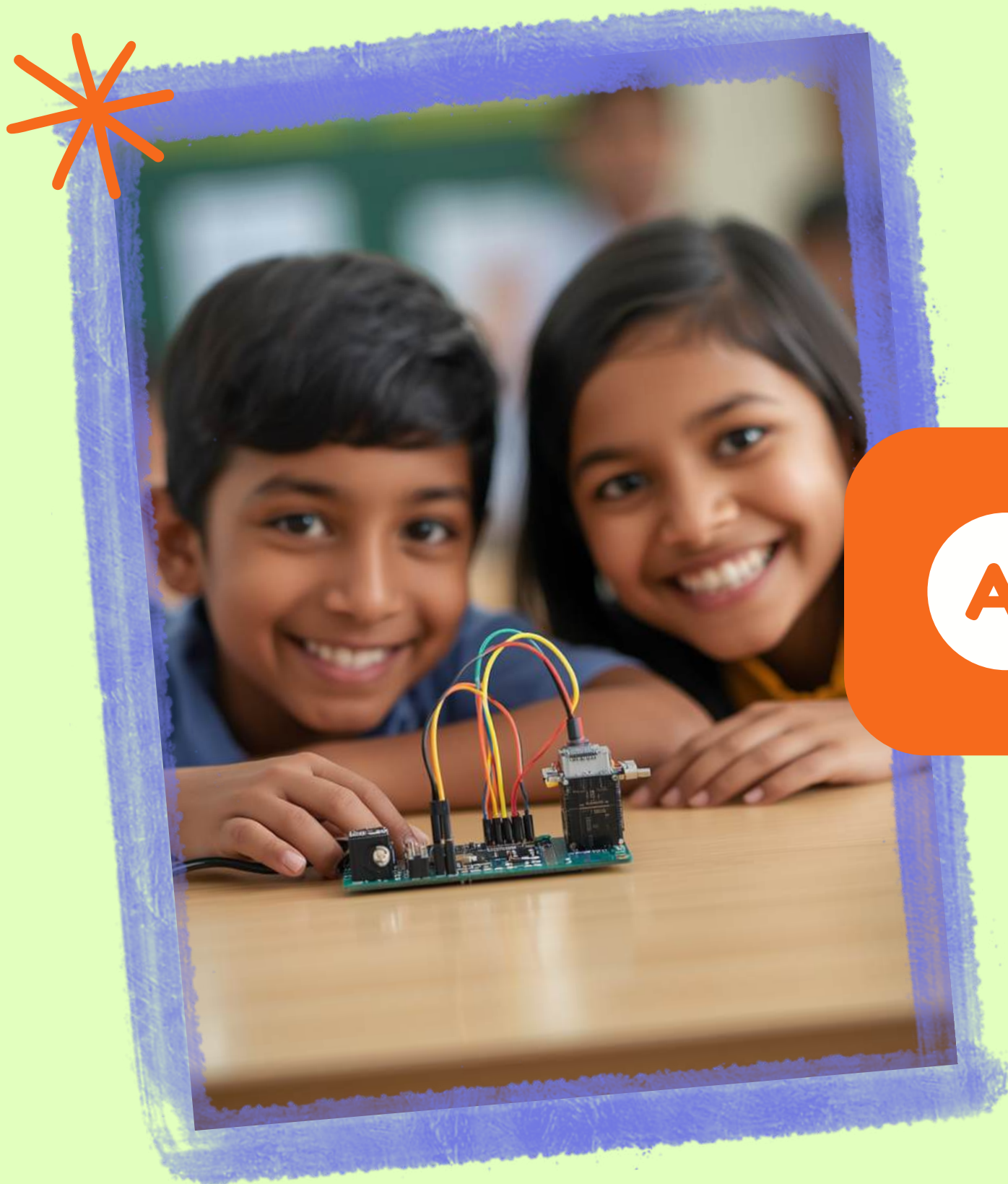
Q2

Which Arduino pin controls the servo signal?

Which digital PWM pin did we use to connect the servo signal wire in today's session? Bonus: What angle opens the gate fully?



Quiz Answers & Explanation



Answers 1&2

A1

A1: SG90 has 3 wires — Red (VCC/Power), Brown or Black (GND/Earth), Yellow or Orange (Signal). A2: Digital PWM Pin 9 on Arduino UNO controls the servo signal.

A3

Answer 3

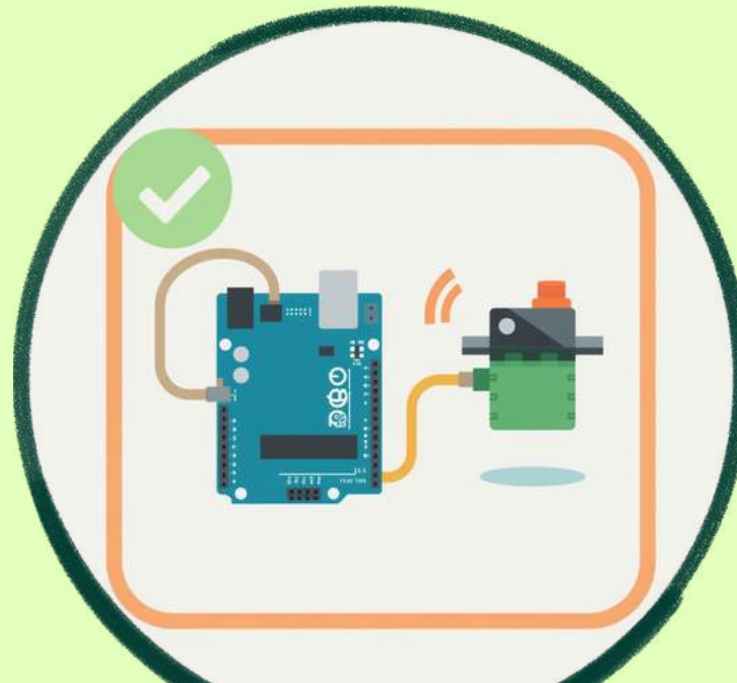
A3: 90 degrees opens the gate fully — just like a real boom barrier at a toll gate or society entrance! The servo holds that exact angle until told to move.

Wrap-Up Thank You!

Great work today! You have learned how servo motors work and how to control them with Arduino to build smart gates. Keep experimenting, keep coding, and get ready for an even more exciting next session!



Today's Learning



Coming Up Next!



Keep Coding!